CHAPTER

16

Non-human DNA

While the vast majority of forensic DNA typing performed for criminal investigations involves human DNA, it is not the only source of DNA that may be useful in demonstrating the guilt or innocence of an individual suspected of a crime (Sensabaugh & Kaye 1998). Recent books have reviewed efforts with non-human DNA (Coyle 2008), wildlife DNA testing (Linacre 2009), and microbial forensics (Budowle et al. 2011).

Domestic animals such as cats and dogs live in human habitats and deposit hair that may be used to place a suspect at the crime scene (D'Andrea et al. 1998). DNA testing can benefit wildlife law enforcement efforts to eliminate poaching or sale of products from endangered species. Demonstration that a botanical specimen came from a particular plant can aid the linkage of a crime to a suspect or help demonstrate that the body of a deceased victim may have been moved from the murder site. DNA testing can be used to link sources of marijuana. A large area of future application for forensic DNA typing involves identification of bioterrorism materials such as anthrax. This chapter will briefly discuss each of these topics and the value of non-human DNA testing in forensic casework. The reference list at the back of the chapter will lead interested readers to additional sources of information on various topics relating to non-human DNA.

As most forensic DNA laboratories will not have the capability or in-house expertise to handle non-human DNA cases when they occasionally occur, samples are often outsourced to specialist laboratories with the appropriate expertise in species identification and animal or plant DNA testing (Ogden 2010). These laboratories may be academic research groups rather than forensic laboratories with established quality assurance measures.

A DNA Commission of the International Society of Forensic Genetics (ISFG) has provided 13 recommendations regarding the use of non-human (animal) DNA in forensic genetic investigations (Linacre et al. 2011; D.N.A. Box 16.1). Recommendations for animal DNA forensic and identity testing have also been made by a group of scientists from the United States, Italy, Germany, Austria, The Netherlands, and Australia who met at the September 2004 International Society for Animal Genetics (ISAG) meeting (Budowle et al. 2005). These recommendations review the importance of using established standard operating procotols (SOPs) along with positive and negative controls. Standardized nomenclatures for loci and alleles are advocated in order to facilitate interlaboratory comparisons and use of

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D.N.A. BOX 16.1

ISFG RECOMMENDATIONS FOR USE OF NON-HUMAN (ANIMAL) DNA

Based on discussions held at the 23rd Congress of the International Society for Forensic Genetics (ISFG) in Buenos Aires, Argentina (September 2009), an ISFG DNA Commission has outlined 13 recommendations for animal DNA testing (Linacre et al. 2011):

1. The same procedures to ensure integrity and traceability of the items should be employed in the collection and examination of animal samples as undertaken for any other forensic investigation.
2. Validation studies from non-domesticated species should use voucher specimens where possible. If this is not possible, then a justification needs to be made for the sample type used.
3. The choice of locus/loci used in species identification, such as, but not restricted to, the mitochondrial genes cyt b, COI, and the D-loop region, needs to be justified based on the ability to identify the unknown species among those that are close genetic relatives.
4. The nucleotide sequence and map showing the location of the primers used in species testing needs to be provided or referenced to a previously published article.
5. Intraspecies and interspecies studies should be provided for any novel primer set used in species identification. The process undertaken to validate the test should be provided, including, but not exclusively, studies on sensitivity, specificity, reproducibility and mixed samples.
6. Primers used to amplify polymorphic DNA should be tested to ensure specificity and reproducibility and should be published in the public domain.
7. If repeat-based polymorphic loci are used for individualization, tetrameric short tandem repeat systems should be used preferentially.
8. Sequenced allelic ladders are essential for the accurate designation of alleles and should be used in all STR typing. The number of repeats should be the basis of reporting of results rather than using only the size based on the number of base pairs of any samples tested.
9. In relationship testing, the mutation probabilities of the STR alleles should be estimated if encountered, or at least the probability of a mutational event occurring should be considered when there is genetic inconsistency at a single or few loci while all other loci show genetic consistency.
10. Relevant population and forensic genetic parameters including allele frequencies should be estimated.
11. A kinship factor should be determined and applied in any calculation. The type of kinship factor applied should be stated clearly and justification should be made for the factor incorporated.
12. A comprehensive casefile should be maintained. A likelihood ratio approach is the recommended way to evaluate the weight of the evidence, considering more than one proposition.
13. Accreditation should be sought if DNA testing of non-human animal DNA for a particular purpose is to become routine.

***Source*:**

Linacre, A., et al. (2011). ISFG: Recommendations regarding the use of non-human (animal) DNA in forensic genetic investigations. Forensic Science International: Genetics (in press). *doi:10.1016/j.fsigen.2010.10.017.*

well-characterized cell lines or reference DNA samples is encouraged to enable compatibility between testing laboratories. Population data should be collected in order to assess allele frequencies and data behind the reference database made available for review so that assumptions involved in the statistical estimate are transparent.

SOPs need to be based on laboratory validation that has defined the operational limits of the technique involved. Scientists using these protocols should successfully complete a qualifying test before performing casework and participate in regular proficiency testing. Casework files should be maintained for each case and should undergo technical and administrative review. Laboratories that do not meet these best practices should disclose to their customer what part of these recommended quality assurance practices are not being met (Budowle et al. 2005).

DOMESTIC ANIMAL DNA TESTING

Budowle et al. (2005) note that genetic analysis with animal DNA samples can help resolve criminal and civil cases as well as aiding kinship analysis with applications such as determining the sire of an offspring when a female has been exposed to multiple males. This chapter will first discuss DNA testing done with domestic animals and then review work with wildlife testing and species identification.

The American Pet Products Association reported in their 2007 and 2008 national pet owners survey that over 71 million U.S. households own a pet (see http://www.appma.org). Their survey found 88 million cats and 75 million dogs in these households, which make up almost two-thirds of all U.S. residences. Since many of these domestic animals shed hair, these hairs could be picked up or left behind at the scene of a crime by a perpetrator. An assailant may unknowingly carry clinging cat hairs from a victim's cat away from the scene of a crime, or hair from the perpetrator's cat may be left at the scene.

The Veterinary Genetics Laboratory (VGL) at the University of California-Davis (see http://www.vgl.ucdavis.edu/forensics) has been performing forensic animal DNA analyses since 1996. The VGL website notes that there are three types of animal DNA evidence: (1) the animal as victim, (2) the animal as perpetrator, and (3) the animal as witness.

Animal abuse cases or the theft of an animal can sometimes be benefited by the power of DNA testing. The remains of a lost pet can be positively identified through genetic analysis. Typically genetic markers like short tandem repeats (STRs) and mitochondrial DNA (mtDNA) are examined in much the same way as with human DNA.

When animals are involved in an attack on a person, DNA typing may be used to identify the animal perpetrator (e.g., a Pit Bull). If the victim is deceased, then DNA evidence may be the only witness that an animal in custody committed the crime. Animal DNA testing can "exonerate" innocent animals so that they are not needlessly destroyed.

Animal DNA has been used successfully to link suspects to crime scenes. A study on the transfer of animal hair during simulated criminal behavior found that hundreds of cat hairs or dog hairs could be transferred from the homes of victims to a burglar or an aggressor (D'Andrea et al. 1998). In fact, the number of hairs found was so high that the authors of this study felt that it is almost impossible to enter a house where a domestic animal lives without being "contaminated" by cat and/or dog hairs even when the owner describes his or her

animal as a poor source of hair (D'Andrea et al. 1998). Due to the fact that shed hairs often do not contain roots, nuclear DNA may not be present in sufficient quantities for STR typing. Mitochondrial DNA may be a more viable alternative for many of these types of shed hair transfers.

Cat DNA

Cats have 18 pairs of autosomes and the sex chromosomes X and Y. Genetic markers have been developed on each of the *Felis catus* chromosomes (Menotti-Raymond et al. 1999). Menotti-Raymond et al. (2005) examined 49 candidate tri- and tetranucleotide STR loci and selected 11 of these as a forensic panel for genetic individualization of domestic cat samples. A multiplex PCR assay utilizing these 11 STRs, which are on nine different chromosomes, has been dubbed the "MeowPlex" (Butler et al. 2002). The power of discrimination with this 11plex feline STR multiplex ranged from 5.5×10^7 to 3.3×10^{13} across the various breeds (Menotti-Raymond et al. 2005).

A gender identification marker was also included in this assay through the addition of PCR primers that are specific for the SRY gene on the cat Y-chromosome. The PCR products for this 12plex amplification fall in the size range of 100 bp to 400 bp and use three dye colors (Figure 16.1). In another version of this assay, two of the loci were moved into a fourth dye color to avoid overlaps in PCR product sizes with the discovery of new alleles (Menotti-Raymond et al. 2005).

Feline STR allele frequencies from domestic cats have been published (Menotti-Raymond et al. 1997a) for the purpose of demonstrating uniqueness of DNA profiles in forensic investigations, such as used in the Beamish case with his cat "Snowball" (Menotti-Raymond et al. 1997b; see also Butler 2010, *Fundamentals*, D.N.A. Box 15.3). The International Cat Association (TICA; see http://www.tica.org/) recognizes 55 breeds of cat.

Population studies on over 1200 cats from 37 different breeds have been conducted by the Laboratory of Genomic Diversity at the National Cancer Institute-Frederick Cancer Research and Development Center in Frederick, Maryland (Menotti-Raymond et al. 2005). In an initial

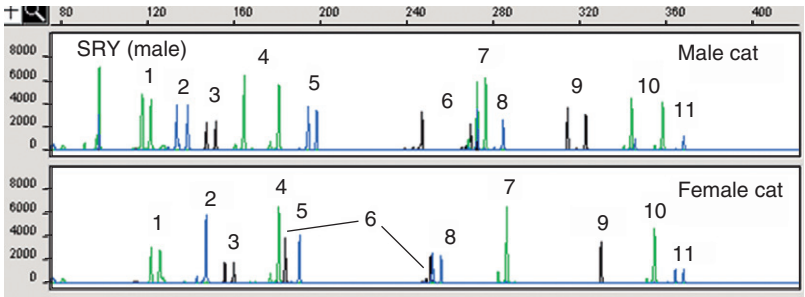


FIGURE 16.1 DNA profiles produced from male (top panel) and female (bottom panel) cat DNA using a multiplex STR typing assay dubbed the "MeowPlex" (Butler et al. 2002). This test examines 11 autosomal STRs and a region of the SRY gene contained on the Y-chromosome that can be used for sex determination.

study of 223 cats from 28 different breeds, the MeowPlex exhibited an average composite locus heterozygosity of 0.73 across the breeds (Menotti-Raymond et al. 2003).

The commonly used cat STR loci are described in Table 16.1. A section of the NIST STRBase website also includes information on cat STRs: http://www.cstl.nist.gov/biotech/strbase/catSTRs.htm.

A real-time quantitative polymerase chain reaction (qPCR) assay (see Chapter 3) for estimating the DNA yield extracted from domestic cat specimens has been developed (Menotti-Raymond et al. 2003). This assay is capable of detecting down to 10 femtograms of feline genomic DNA and uses high-copy number short interspersed nuclear elements (SINEs) similar to the *Alu* repeats described in Chapter 12.

TABLE 16.1 Commonly Used Cat STR Loci. The 11 Loci Shaded in Gray Are Part of a Proposed Forensic Typing System (Menotti-Raymond et al. 2005) Co-Amplified as the "MeowPlex" (Butler et al. 2002). The Additional Dinucleotide Loci and FCA 441 Make up a Core Panel for Cat Identification and Parentage Testing Proposed by the International Society for Animal Genetics (Lipinski et al. 2007).

Locus	Cat Chromosome	Repeat Motif	Size Range	Reference
FCA 733	B2	GATA	128–226 bp	Menotti-Raymond et al. (2005)
FCA 723	A1	$(GGAA)_nG(GAAA)_m$	243–317 bp	Menotti-Raymond et al. (2005)
FCA 731	B1	CCAT	337–401 bp	Menotti-Raymond et al. (2005)
FCA 736	B4	$(ATAC)_n\ (CA)_3\ (ATAG)_m$	164–222 bp	Menotti-Raymond et al. (2005)
F 124	E1	GAAA	225–367 bp	Menotti-Raymond et al. (2005)
FCA 742	D4	CTTT	122–175 bp	Menotti-Raymond et al. (2005)
F85	B1	$(CTTT)_n\ (CT)_{10}\ (T)_4\ (CTTT)_m$	183–301 bp	Menotti-Raymond et al. (2005)
FCA 740	C1	GATA	308–336 bp	Menotti-Raymond et al. (2005)
F 53	A1	GAAA	115–272 bp	Menotti-Raymond et al. (2005)
FCA 749	F2	GATA	276–416 bp	Menotti-Raymond et al. (2005)
FCA 441	D3	TAGA	113–137 bp	Menotti-Raymond et al. (2005)
			133–173 bp	Lipinski et al. (2007)
FCA 069	B4	AC	88–116 bp	Lipinski et al. (2007)
FCA 075	E2	TG	112–146 bp	Lipinski et al. (2007)
FCA 105	A2	TG	173–207 bp	Lipinski et al. (2007)
FCA 149	B1	TG	120–136 bp	Lipinski et al. (2007)
FCA 220	F2	CA	208–224 bp	Lipinski et al. (2007)
FCA 229	A1	GT	150–174 bp	Lipinski et al. (2007)
FCA 310	C2	$(CA)_n\ TA\ (CA)_m\ TA\ (CA)_p$	112–138 bp	Lipinski et al. (2007)
FCA 678	A1	AC	222–236 bp	Lipinski et al. (2007)

Dog DNA

While cat DNA testing may be involved in situations where the animal hair acts as a silent witness to connecting a perpetrator to a crime scene, evidence from dogs is more frequently linked to situations where the animal is the perpetrator. Rottweilers, German Shepherds, Doberman Pinschers, and Pit Bulls can be trained as security animals and may attack, injure, or even kill people. For example, with a canine population in Australia of around 4 million, there are an estimated 100,000 dog attacks each year—and many of them go unsolved (Clarke & Vandenberg 2010).

Canine STR Loci and Assays

A number of STR markers have been mapped and characterized on the 38 pairs of autosomes and the X-chromosome of *Canis familiaris*, the domestic dog (Neff et al. 1999). Early dog STR assays included many dinucleotide repeat loci but tetranucleotide loci with lower stutter have been the focus of more recent efforts. However, different groups have targeted different sets of loci with almost no overlap between them (Halverson & Basten 2005, Berger et al. 2008, van Asch et al. 2009, Tom et al. 2010). There is a need for standardization on the loci and allele nomenclatures used (Berger et al. 2009). In an early effort, 15 canine STR loci were characterized with sequenced alleles (Eichmann et al. 2004).

The commonly used canine STR loci are described in Table 16.2. A section of the NIST STRBase website also includes information on dog STRs: http://www.cstl.nist.gov/biotech/strbase/dogSTRs.htm. The PEZ locus names come from Perklin-Elmer Zoogen, a company that developed the StockMark kits for Applied Biosystems back in the mid-to-late 1990s. The FH locus names came from the Fred Hutchinson Cancer Research Center (Seattle, Washington).

Finnzymes (Espoo, Finland) has also released a commercial kit with 18 STR loci and amelogenin for sex-typing that utilizes dinucleotide markers (see http://diagnostics.finnzymes.fi/canine_genotypes.html).

A set of 10 dinucleotide repeat STRs has been used to aid investigations in illegal animal deaths (Padar et al. 2001) and a dog attack that resulted in the death of a seven-year-old boy (Padar et al. 2002). In addition, it was demonstrated that DNA profiling of human blood recovered from a dog's fur can associate or exonerate the animal from connection to an attack (Brauner et al. 2001).

Canine mtDNA Testing

Canine mitochondrial DNA possesses two hypervariable regions (HV1 and HV2) similar to the human mtDNA described in Chapter 14. Savolainen et al. (1997) found 19 sequence variants across a 257 bp segment of the hypervariable region 1 of the mtDNA control region in 102 domestic dogs of 52 different breeds. They concluded that on average 88 out of 100 tested animals could be excluded with this mtDNA sequence analysis. By way of comparison in 100 British white Caucasians an exclusion capacity of 0.97 was observed (Piercy et al. 1993). While domesticated dog mtDNA is not as variable as human mtDNA, it can still provide helpful clues in forensic cases (Savolainen and Lundeberg 1999, Schneider et al. 1999).

TABLE 16.2 Information on Some Commonly Used Dog STR Loci.

Locus	Canine Chromosome	Repeat Motif	Size Range	Reference
PEZ01	7	TACA	95–136 bp	StockMarks kit
PEZ02	17	GGAA	104–144 bp	Tom et al. (2010)
PEZ03	19	GAA	95–154 bp	StockMarks kit
PEZ05	12	TTTA	92–116 bp	Tom et al. (2010); StockMarks kit
PEZ15	16	AGAA	183–249 bp	Berger et al. (2008)
PEZ16	27	GAAA	281–332 bp	Tom et al. (2010)
PEZ17	4	GAAA	191–225 bp	Tom et al. (2010)
PEZ20	22	AAAT	152–202 bp	StockMarks kit
PEZ21	2	AAAT	83–103 bp	Tom et al. (2010)
FH2001	23	GATA	119–160 bp	Tom et al. (2010)
FH2004	11	AAAG	233–325 bp	Tom et al. (2010)
			161–257 bp	van Asch et al. (2008)
FH2010	24	ATGA TTCA	222–243 bp	Tom et al. (2010)
			154–170 bp	van Asch et al. (2008); StockMarks kit;
FH2017	15	AGGT/AGAT/GATA	257–276 bp	Tom et al. (2010)
FH2054	12	GATA	139–177 bp	Tom et al. (2010); Finnzymes kit StockMarks kit
FH2088	15	TTTA/TTCA	94–138 bp	Tom et al. (2010)
FH2107	3	GAAA	292–426 bp	Tom et al. (2010)
FH2309	1	GAAA	340–428 bp	Tom et al. (2010)
FH2328	33	GAAA	171–213 bp	Tom et al. (2010)
FH2361	29	GAAA TTTC	323–439	Tom et al. (2010)
			231–347 bp	van Asch et al. (2008)
FH2079	24	GGAT	263–299	Berger et al. (2008); StockMarks kit
FH3313	19	GAAA	341–446 bp	Tom et al. (2010)
FH3377	3	GAAAA	183–305 bp	Tom et al. (2010)
FH2658	14	GAAA	106–138 bp	van Asch et al. (2008)
FH3210	2	AAGA	230–315 bp	van Asch et al. (2008)
FH3241	8	TTCT	250–270 bp	van Asch et al. (2008)
FH4012	15	TTTC	119–143 bp	van Asch et al. (2008)
REN214L11	16	GAAT	154–162 bp	van Asch et al. (2008)
C38	38	TTCT	132–217 bp	van Asch et al. (2008)
VWF.X	27	AGGAAT	151–187 bp	Tom et al. (2010)

Efforts have been made to standardize the nomenclature for the canine mtDNA control region (Pereira et al. 2004) and informative sequence variants outside of the control region have also been identified (Webb & Allard 2009a, 2009b).

Other Domesticated Animals

While DNA testing of household animals like dogs and cats can help solve crimes, other domesticated animals—particularly animals used for recreation or sources of food—may be DNA tested for identification purposes. DNA tests have been developed for horses (Dimsoski 2003, van de Goor et al. 2010, Chen 2010), cattle (van de Goor et al. 2009), pigs (Robino et al. 2008), and sheep (Heaton et al. 2010). These genetic identification tests can be used to track the source of tainted meat products such as those obtained from cattle suffering from "mad cow disease." Horse DNA testing can be important for confirming genetic pedigrees and is required for registering some breeds including American Quarterhorses and many racehorses (Bowling et al. 1997, Tozaki et al. 2001).

Sources for Performing Non-Human DNA Testing

Testing of non-human DNA samples is not routinely performed in public forensic laboratories and thus these types of studies have to be outsourced to academic labs or speciality laboratories that focus on testing specific species (see Ogden 2010). As noted earlier, the Veterinary Genetics Laboratory (VGL) at the University of California-Davis has been performing forensic animal DNA analyses since 1996. As of October 2010, the VGL offers genetic analyses for parentage verfication, genetic disease screening, and diagnostic testing on alpaca, beefalo, bison, cat, cattle, deer, dog, elk, goat, horse, llama, pig, sheep, water buffalo, and yak samples.

The U.S. Fish and Wildlife Service Forensic Laboratory (http://www.lab.fws.gov/) in Ashland, Oregon conducts species identification as well as other DNA testing to aid fish and wildlife forensic investigations. Feline STRs and mtDNA testing is performed by QuestGen Forensics (http://www.questgen.biz/), which also does canine STR and mtDNA testing to aid forensic investigations.

SPECIES IDENTIFICATION

There is value in species identification particularly when bones or bone fragments are being uncovered from a gravesite. Are they human remains or those of an animal? Because mtDNA with its higher copy number survives better than nuclear DNA in highly processed or degraded bone or tissue, it has value in species identification. Several informative segments of mtDNA are conserved enough that universal primers can be employed to amplify a target region with subsequent sequence analysis to differentiate between species (Linacre & Tobe 2009).

Typical steps in species identification involve: (1) sequence analysis of a selected variable region of DNA with conserved primers and (2) comparison of these sequence results to a database of sequences such as GenBank usually through phylogenetic analysis to place the seqeuence obtained in context of other previously reported DNA sequences for the same region.

mtDNA Cytochrome b Gene

Sequence analysis of the mtDNA cytochrome b gene is effective at identifying the species of origin for a biological sample (Parson et al. 2000, Hsieh et al. 2001, Branicki et al. 2003, Linacre et al. 2004). Some tests have even been developed to simultaneously amplify both the human mtDNA control and the cytochrome b gene to enable simultaneous human and species identification (Bataille et al. 1999).

mtDNA 12S rRNA Gene

Another mtDNA region examined for species identification is the 12S ribosomal RNA gene (Balitzki-Korte et al. 2005, Melton & Holland 2007). In one study, DNA sequence analysis of an ≈150 bp fragment of the 12S rRNA mtDNA successfully identified dog, feral pig, raccoon, cat, goat, sheep, rat, and yak DNA samples in 12 different forensic cases (Melton & Holland 2007).

mtDNA COI Gene

An international effort, known as the Barcode of Life Project (Ratnasingham & Hebert 2007), is working to catalog different species through utilizing a 648 bp region from the mitochondrial cytochrome c oxidase 1 (COI) gene. The COI gene has proved effective for animals but the mutation rate is too slow to be useful in resolving various plant species from one another. The barcoding gene COI has been explored for use in a forensic context (Dawnay et al. 2007, Wilson-Wilde et al. 2010).

Application for Body Fluid Identification

Species identification techniques can also help with identification of body fluids. Researchers from New Zealand (Fleming & Harbison 2010) used the 16S-23S rRNA intergenic spacer region to help identify vaginal specific bacteria—*Lactobacillus crispatus* and *Lactobacillus gasseri*. They then incorporated their *Lactobacilli* markers into an 11plex mRNA multiplex system to enable identification of circulatory blood, menstrual blood, saliva, semen, and vaginal secretions.

Detection of Streptococcus bacteria, which are only found in the human mouth and saliva, has been used for saliva identification (Nakanishi et al. 2009) and distinguishing bloodspatter originating from an individual's oral cavity (i.e., mouth or nose) from other types of bloodstains (Donaldson et al. 2010).

WILDLIFE DNA TESTING

Interpol estimates that illegal trade of wildlife takes place in a growing global black market at a pace of approximately $20 billion per year (Wilson-Wilde 2010, Alacs 2010). The remains of stolen animals or illegally procured meat (e.g., endangered species or poaching)

can be identified through DNA testing (Giovambattista et al. 2001, Poetsch et al. 2001). Genetic testing can be used to help prosecute individuals who exploit exotic or endangered animals. The U.S. Fish and Wildlife Service's forensic laboratory in Ashland, Oregon does some species identification using DNA (see http://www.lab.fws.gov). Other laboratories performing similar wildlife DNA testing are located in Canada and the UK.

As with any DNA testing method applied to forensic investigations, validation experiments are important to defining assay performance (Dawnay et al. 2008). Allele frequencies are measured to help underpin statistical calculations on the rarity of DNA profiles (Jobin et al. 2008). The small sample sizes collected in some wildlife studies can impact the minimum allele frequencies used.

Techniques for Assessing Genetic Differences

Several techniques may be utilized for assessing genetic differences between organisms being tested. Which technique is best to use will generally depend on the amount of prior information available. Advantages and disadvantages of each approach are compared in Table 16.3.

DNA Sequencing

DNA sequence analysis of specific genes, such as mtDNA cytochrome b, is beneficial in that complete details down to the molecular level are available for the tested specimen. However, sequence analysis is relatively expensive and time-consuming with mostly redundant information. Sample mixtures and multi-ploidy genomes are also more difficult to decipher from sequence information.

TABLE 16.3 Table of Advantages and Disadvantages of Each Approach (See Linacre & Tobe 2009, pp. 68–69).

Approach	Advantages	Disadvantages
Sequence Analysis (universal primers)	Can analyze a wide range of species if using universal primers	Cannot separate mixtures
	A database can be built	
STR or SNP Typing (species-specific primers)	Able to separate mixtures	Prior sequence knowledge is needed to design primers
	May be possible to add new species to a multiplex assay	
RAPD	Inexpensive	Difficult to reproduce within and between laboratories
		Cannot produce a database
AFLP/T-RFLP	Fairly inexpensive	May not have the individual resolving power of STR typing
	A database can be built	

STR Typing

If a specific locus with many alleles can be determined, PCR primers can be designed and allele sizes differentiated. However, the genome sequences for many organisms of interest (plant and animal) are not yet available and STR loci have not been characterized. For this reason, other techniques are sometimes used to help screen for sample genetic differences and similarities.

RAPD

Randomly amplified polymorphic DNA, or RAPD, marker analysis utilizes short PCR primers consisting of random sequences usually in the size range of 8 to 15 nucleotides in length. Complex patterns of PCR products are generated as these random sequence primers anneal to various regions in an organism's genome. RAPD suffers from poor reproducibility between laboratories largely because of the requirement of consistent PCR amplification conditions including thermal cycler ramp speeds. The complex patterns of RAPD also prevent mixture interpretation and provide challenges in consistent scoring of electrophoretic images even in single-source samples.

AFLP

Patterns from amplified fragment length polymorphism markers can be generated with greater reproducibility compared to RAPDs. AFLPs are generated by first cutting a double-stranded DNA sample with one or more restriction enzymes (Vos et al. 1995, Ranamukhaarachchi et al. 2000). Specific "adaptor" sequences are then ligated to the restriction cut sites. PCR primers that recognize these ligated adaptor sequences are used to amplify different sized DNA fragments that can then be separated using electrophoresis. The final result is a complex series of peaks usually in the 50 bp to 400 bp size range that can be scored with computer software and compared with other AFLP patterns.

T-RFLP

In terminal restriction fragment length polymorphism (T-RFLP), PCR is used to amplify a region of bacterial DNA that contains conserved primer binding sites. One primer has a 5′-fluorescent tag to enable detection. Following PCR, the amplicons are digested with a restriction enzyme. Based on sequence differences within the amplified target sequence, different-sized products will be generated. Capillary electrophoresis separation and detection of the different fragments results in a bacterial T-RFLP profile. Profile patterns can be compared with multi-dimensional scaling (MDS) plots (Lenz & Foran 2010).

PLANT & POLLEN DNA TESTING

In the area of plant DNA testing, there are primarily two areas currently being investigated. The first is the linking of plant material to suspects or victims in order to make an association with a particular area where a crime was committed. The second is in linking marijuana to aid in forensic drug investigations. Some of the applications of forensic botany have been reviewed by Heather Miller-Coyle (Miller-Coyle et al. 2001, Coyle 2008).

Linking Plant Materials to Suspects

Crimes often occur in localized areas containing a unique combination of botanical growth. If these plants, algae, or grass are sufficiently rare, then recovery of trace evidence from the clothing of a victim or the personal property of a suspect may be helpful in making an association that can link them to a crime scene (Szibor et al. 1998, Norris and Bock 2000, Horrocks and Walsh 2001).

Although plant DNA testing is not yet used routinely (Bock and Norris 1997), it has helped link suspects to crime scenes and aided important investigations. In the first use of forensic botanical evidence, two small seedpods from an Arizona Palo Verde tree found in the back of a pick-up truck were used to place an accused murderer at the crime scene (Yoon 1993). Genetic testing on the seeds showed that in a "lineup" of 12 Palo Verde trees near the crime scene, DNA from the seeds matched only the tree under which the victim's body had been found. In *State v. Bogan*, the jury found the accused guilty based in large measure on the plant DNA evidence.

Marijuana DNA

Several DNA tests have been developed for *Cannabis sativa* (marijuana) because it is an illegal substance associated with many crime scenes. Marijuana is one of the most commonly identified drugs tested by U.S. forensic laboratories (see http://www.deadiversion.usdoj.gov/nflis).

Marijuana DNA testing can link an individual to a sample, link growers, and help track distribution networks (Miller-Coyle et al. 2001). However, it is important to keep in mind that if the marijuana plants were propagated clonally rather than by seed, then they will have identical DNA profiles. Clonal propagation in marijuana is performed by taking cuttings from a "mother" plant and rooting them directly in the soil to create large numbers of plants having identical DNA (Miller-Coyle et al. 2001).

Efficient extraction protocols have been developed that yield 125 ng to 500 ng of DNA per 100 mg of fresh plant tissue (Miller-Coyle et al. 2003a). DNA testing of marijuana as with other plants has traditionally been performed with one of three methodologies: randomly amplified polymorphic DNA markers (RAPDs), amplified fragment length polymorphisms (AFLPs), or short tandem repeats (STRs). These techniques and their specific application to marijuana DNA typing have been reviewed (Miller-Coyle et al. 2003b). Even highly inbred individual plants can be distinguished by their AFLP patterns (Miller-Coyle et al. 2003b).

A few STR markers have been characterized in *Cannabis sativa* and developed into effective DNA tests (Hsieh et al. 2003, Gilmore et al. 2003, Alghanim & Almirall 2003). As with human STRs, marijuana STR markers are highly polymorphic, specific to unique sites in the genome, and capable of deciphering mixtures. A hexanucleotide repeat marker showed repeat units ranging from 3 to 40 in 108 tested marijuana samples, and primers amplifying this locus produced no cross-reactive amplicons from 20 other species of plants tested (Hsieh et al. 2003).

Efforts have been made to improve molecular techniques for identifying marijuana plants and to develop comparative databases to serve as effective tools for law enforcement

purposes. In order to determine the possibility of a random match with marijuana seizure samples, it is important to have a database of seizure samples so their DNA profiles can be used for comparison (Miller-Coyle et al. 2003).

Bacterial DNA in Soil

Soil is filled with a variety of microscopic organisms including bacteria and fungi. It has been shown that different combinations of bacteria in soil samples from different locations can be differentiated with DNA testing (Horswell et al. 2002, Heath & Saunders 2008, Lenz & Foran 2010).

The use of these DNA signatures from soil microbes are being explored to see if this information can be used to link individuals to a specific location based on soil found on their shoes or clothing. This type of forensic soil evaluation could then potentially link a suspect to a crime scene or identify where a victim was killed and whether the body was moved to a different location. Appropriate sampling still remains a challenge as well as overcoming PCR inhibitors present in soil, but this is an area that could bear fruit in the future.

IDENTIFICATION THROUGH HUMAN PARASITES

Research has demonstrated that it is possible to differentiate between individuals based on DNA sequence differences found within parasites, such as viruses found within the individuals. Japanese researchers have focused on various methods of typing the JC human polyomavirus as a means for tracing geographic origins of unidentified cadavers (Ikegaya et al. 2002, Ikegaya et al. 2008, Ikegaya 2008).

MICROBIAL FORENSICS

Unfortunately microbial forensics will likely become a larger part of DNA testing in the future with the threat of terrorism and the use of biological warfare agents. Microbial evidence can be obtained from either real terrorist events or hoaxes. The efforts in this area will likely require forensic laboratories to build strong collaborations with academia, private sector, and national laboratories. Important requirements of bio-threat detection assays are high sensitivity, high specificity in complex samples, fast measurement, compact design for portability and field use, and internal calibration and reference to ensure reliable results (Ivnitski et al. 2003).

In October 2001 a bioterrorism attack impacted the United States as government offices and media outlets received anthrax-laden letters sent anonymously through the postal service. This attack resulted in 22 anthrax cases and five deaths. In addition, many people were afraid to open their mail for months afterwards. In the two years following this attack, more than 125,000 samples were processed as part of this case (Popovic & Glass 2003). In August 2008, almost seven years after the anthrax attack, the FBI Laboratory announced a

breakthrough in the "Amerithrax" investigation with plans to charge Dr. Bruce Ivins, a scientist at the U.S. Army Medical Research Institute for Infectious Diseases (USAMRIID) in Frederick, Maryland. Dr. Ivins committed suicide before the charges could be filed.

Several challenges arise when trying to gather evidence, identify the bio-crime organism(s), and trace the source of the organism(s). First responders to crime scenes where biological weapons have been dispersed have to be concerned about their own safety and the safety of others while maintaining chain of custody of any evidence collected from the crime scene, all the while trying to prevent contamination of the evidence and the environment. Databases need to be established for intrinsic background species and bio-threat strains. Reliable reference material is needed for comparison purposes. Proficiency and validation testing are necessary to estimate false-positive and false-negative rates (Kiem 2003).

The U.S. efforts in building a response to bioterrorism have been announced in a policy paper (Budowle et al. 2003). Shortly after the attacks of September 11, 2001, the FBI initiated a Scientific Working Group on Microbial Genetics and Forensics (SWGMGF) that helped develop guidelines related to the operation of microbial forensics (SWGMGF 2003). Currently there are an insufficient number of validated analytical tools to characterize and identify biological agents that might be used in a terrorist attack (Budowle 2003). Research efforts will continue in this area.

Comparative genome sequencing promises to be a powerful tool for investigating infectious disease outbreaks as was performed with the whole-genome sequencing of *Bacillus anthracis* (anthrax) (Read et al. 2002a, 2002b). Phylogenetic analyses of viral strains of human immunodeficiency virus (HIV) have been admitted and used as evidence in court (Metzker et al. 2002, Scaduto et al. 2010). However, since bacteria and viruses reproduce asexually, clones are prevalent. A perfect match between evidence collected and a reference sample is much less definitive than with human identity testing where sexual reproduction shuffles genetic material each generation.

For more information on microbial forensics, a reference book is now in its second edition and should be consulted by those interested in this topic (Budowle et al. 2011).

CHALLENGES WITH PRESENTING NON-HUMAN DNA IN COURT

In a 1998 article, George Sensabaugh and David Kaye considered several issues regarding whether a given application with non-human DNA is ready for court use (Sensabaugh & Kaye 1998). These issues include the novelty of the application, the validity of the underlying scientific theory, the validity of any statistical interpretations, and the existence of a relevant scientific community to consult in assessing the application.

Many times new methods are applied for the first time in microbial forensics or animal or plant DNA testing that have not yet undergone the scrutiny of regular forensic DNA testing techniques. Reference DNA databases for comparison purposes and use in calculating the probability of a chance match take time to develop and may not be in place prior to an investigation. Finding appropriate experts to review the scientific soundness of a novel application can also be challenging. Issues surrounding outsourcing of sample testing and appropriate validation will always be a challenge in this area. Nevertheless, the power

and influence of forensic DNA testing will continue to grow as it is used in more and more diverse applications to solve crimes that were previously inaccessible.

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